Q. Ai

QuantSpace.Ai

Building a comprehensive AI app surpassing Perplexity, ChatGPT, Gemini, and Copilot is feasible for a solo developer by chaining free/paid APIs for LLMs, search, images, and more—leveraging your QuantAI experience for a niche edge like finance/trading integration. Costs start near-zero on free tiers but scale to $50-500/month at moderate usage (10K queries/day), depending on traffic and premium models.

Free APIs

These have unlimited or generous no-cost tiers for prototyping your MVP.

- **LLMs**: Google Gemini 1.5 Flash (15 RPM, 1M tokens/min), Hugging Face Inference (unlimited public models like Llama 3 8B), Groq free tier (Llama/Mistral models), Cloudflare Workers AI (Llama 3 variants).
- **Web Search**: Search1API (no-key instant access), DuckDuckGo Instant Answer, Tavily Researcher (1,000 credits/month).
- **Image Generation**: Puter.js (unlimited DALL-E/Flux/Stable Diffusion via frontend), Pixazo free credits (Flux/SDXL).
- **Vector DB**: Open-source self-hosted like Valkey, Faiss (local via Flutter plugins—no cloud cost).
- **Other**: NewsAPI.org (headlines), FRED API (free economic data for your finance focus).

Semi-Free APIs

Free tiers for low usage, pay-per-use beyond limits—ideal for scaling QuantAI-style apps.

API	Free Tier	Paid Beyond	Best For
Tavily Search	1,000 credits/mo	$30/mo for 4,000	AI-optimized SERPs

API	Free Tier	Paid Beyond	Best For
SerpAPI	Trial credits	$25/1K searches	Google/Bing scraping
Exa	Trial	$5/1K neural search	Semantic web research
Groq	Free credits	$0.20/1M tokens (Gemma)	Fast inference
Hugging Face Spaces	CPU hosting	GPU grants/pro	Custom models

Paid APIs

Premium for top performance; use for "better than" edge via advanced reasoning/multimodal.

Provider/Model	Input $/1M Tokens	Output $/1M Tokens	Notes
OpenAI GPT-5.2 Pro	$21	$168	Versatile, high-context
Anthropic Claude 4.5 Opus	$5	$25	Reasoning/coding
Google Gemini 3 Pro	$0.10-$2	Varies	Multimodal free tier upgrade
Mistral (via providers)	$0.20-$3	$1-$15	Efficient Euro models

Cost Estimates

Prototype (100 queries/day): $0 (Gemini + Search1API + Puter.js).

Growth (10K queries/day, mixed free/paid): $50-200/mo (Tavily $30 + Groq/Gemini $20 + hosting).

Scale (100K users/mo, premium): $500-2K/mo (OpenAI/Claude $300 + SerpAPI $200 + Vercel $100); offset via ₹999/semester subs like your QuantAI plan.

Tips for You: Start Flutter MVP with Gemini + Tavily (free tiers cover college launch), add RAG via local Faiss for notes/finance PDFs. Deploy Firebase (free tier) backend. Your prior QuantAI codebases speed this to 1 week. Track via Helicone for optimization.